Cell-free reconstitution of PENK proteolytic processing

Purpose. Reproduce, in a tube, the proteolytic cascade that converts proenkephalin into its opioid peptide products — and determine whether the engineered construct (engPENK) is processed equivalently to the native precursor (P01210).

Design. Two stages, two constructs, run as paired arms:

	Stage 1	**Stage 2**
Substrate	24 synthetic junction peptides (16-mer)	full-length recombinant proprotein
Purpose	establish enzyme conditions; measure per-site cleavage	confirm in conformational context
Readout	LC-MS/MS, 7 multiplexed pools	LC-MS/MS, intact + peptide-level

The **native arm is the built-in positive control**: every native product is documented in the literature, so a native reaction that fails to produce Met-enkephalin invalidates the run before any engPENK result is interpreted.

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1. Site architecture (computed from the sequences)

Both constructs carry **7 enkephalin cassettes** and an identical junction-tier architecture — 10 canonical dibasic (KR/RR), 2 slow dibasic (KK), 4 licensed monobasic — offset only by the cassette-length difference (native 267 aa, engPENK 288 aa).

Two engineered cassettes will **not** give a clean uniform product:

- **Cassette 4** runs into `RGLKRSPQ` — the native Met-enk-Arg-Gly-Leu extension is retained.
- **Cassette 7** ends in `RF` at the C-terminus — no downstream dibasic exists.

Two divergences that this experiment is designed to detect

(a) Amidation. 5 of 7 engineered cassettes terminate in Gly (`YGGFEVVG`) and are therefore **PAM substrates**. **No native cassette is** (0 of 7). If PAM is active, engPENK yields `YGGFEVV-NH₂` (769.3879, $[M+H]^+$) — a species with no native counterpart.

(b) Cathepsin L accessibility. The two routes cut in different places, which is why the cassette swap affects them unequally [Yoon2022]:

- **PC1/3 and PC2** cleave on the **C-terminal side** of the dibasic.
- **Cathepsin L and V** cleave on the **N-terminal side of, and between**, the dibasic, with a preference for hydrophobic residues at P2 — specifically **Leu, Trp, Phe, Tyr**.

For the **convertase route the scissile context is unchanged** by the swap: P1 and P1′ are identical between constructs at all seven cassettes (the dibasic itself was not altered).

For the **cathepsin route the P2 residue changes at all seven sites**:

	native	**engPENK**
cassettes 1-5, 7	`YGGF·M↓KR/KK/RR` — P2 = **Phe**, P1 = Met	`YGGFEVV·G↓KR/KK/RR` — P2 = **Val**, P1 = Gly
cassette 6	`YGGF·L↓KR` — P2 = **Phe**, P1 = Leu	`YGGFEVV·G↓KR` — P2 = **Val**, P1 = Gly
sites with P2 in the preferred set (Leu/Trp/Phe/Tyr)	**7 of 7**	**0 of 7**

Native P2 is Phe at every cassette — one of the four residues the source names explicitly. engPENK P2 is Val at every cassette: still hydrophobic, but **not** among the four preferred residues. P1 additionally changes from Met/Leu to Gly.

Prediction: engPENK is a poorer cathepsin L substrate than native PENK, while remaining an equivalent convertase substrate. This is a direct, falsifiable consequence of the cassette design and is the single most informative thing Stage 1 can tell you. It is not a defect of the construct if the convertase route dominates in the target cell — but it must be measured, not assumed. Note the prediction is directional, not quantitative: Val at P2 is a demotion from the preferred set, not an abolition, so expect reduced rate rather than no cleavage.

Figure 1 — Junction map. Every predicted endoproteolytic site in both constructs, coloured by tier. Cassette positions are shaded; numbers above the bar index the seven enkephalin cassettes.

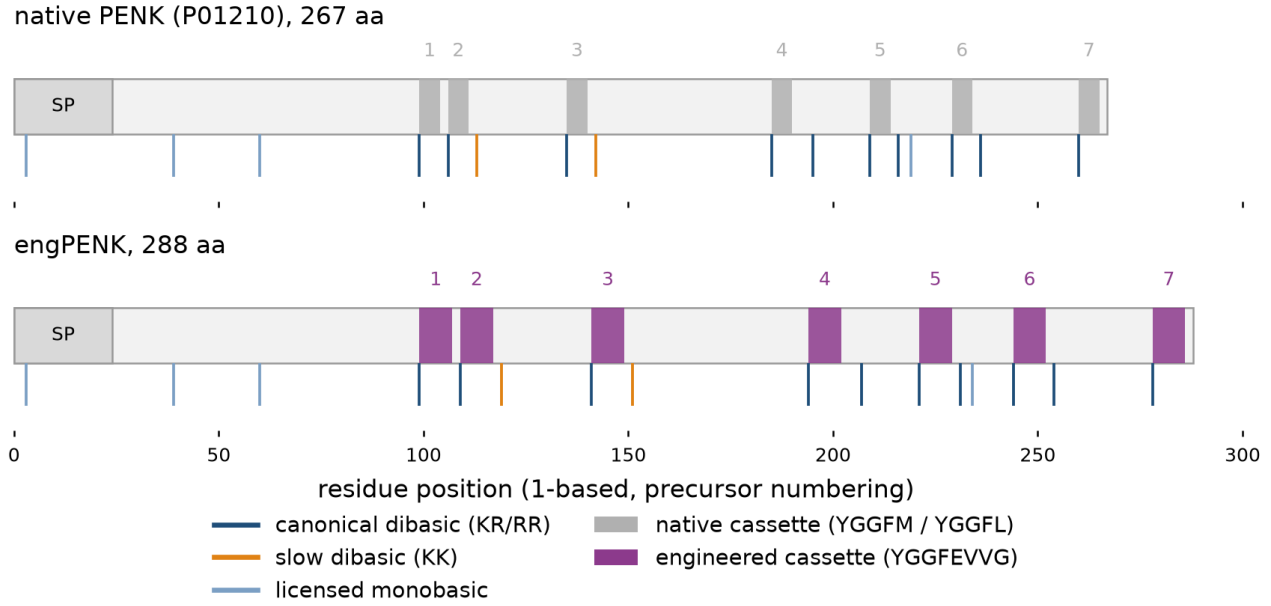
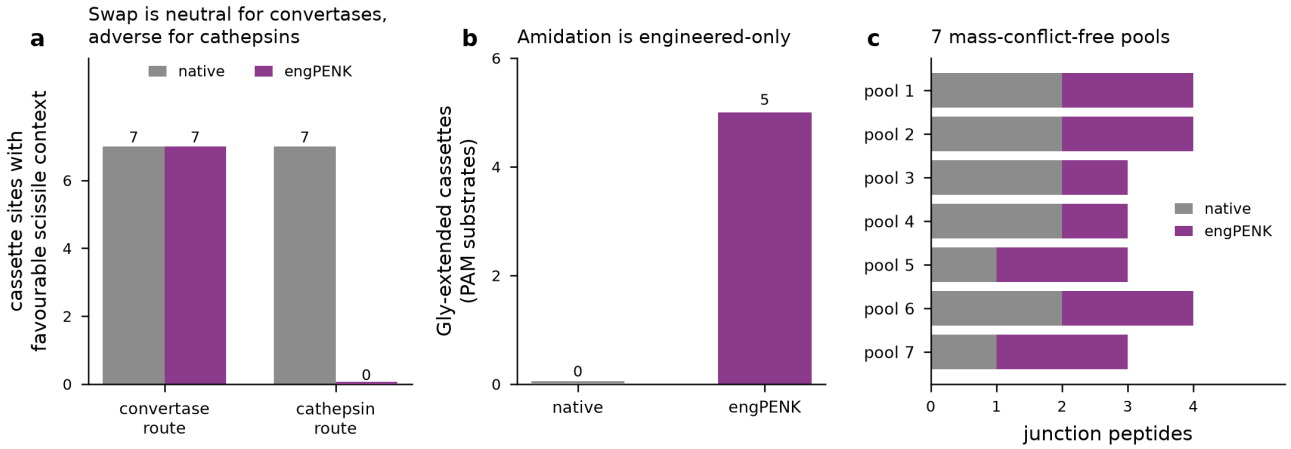


Figure 2 — Design summary. (a) The cassette swap leaves the convertase scissile context unchanged at all seven sites but removes the preferred cathepsin L P2 residue at all seven. (b) Gly-extended, PAM-competent cassettes exist only in engPENK. (c) The 24 junction peptides split into 7 mass-conflict-free pools.



2. Enzymes, buffers, and cofactors

All buffer compositions below marked **[verbatim]** are quoted from the cited primary source. Values from database annotation are marked **[UniProt]**.

2.1 PC1/3 (PCSK1) — [verbatim Yoon2022]

- 50 mM Na-acetate, **pH 5.5**
- 5 mM $CaCl_2$
- 0.1% octyl-β-glucopyranoside (bOG)
- 37 °C
- Published working concentration: **3.6 ng/µL**
- Cofactor: Ca^{2+} [UniProt P29120]

2.2 PC2 (PCSK2) — [verbatim Yoon2022]

- 50 mM Na-acetate, **pH 5.5**
- 5 mM $CaCl_2$
- 0.1% bOG
- **plus 100 mM NaCl** (PC2 only)
- 37 °C
- Published working concentration: **5.4 ng/µL**
- **Requires the 7B2 chaperone (SCG5)** for maturation and activity [UniProt P05408]. Source PC2 as the 7B2-co-expressed, pre-activated enzyme; do not assume recombinant PC2 alone is active.

2.3 Cathepsin L (CTSL) — [verbatim Yoon2022]

- 50 mM citrate phosphate, **pH 5.5**
- 1 mM EDTA
- 100 mM NaCl
- 5 mM DTT
- 37 °C
- Published working concentration: **0.04 ng/µL**
- **No preactivation required** (cathepsin V does require it: 5 mM DTT, 20 mM Na-acetate pH 5.5, 1 mM EDTA, 100 mM NaCl, 37 °C, 30 min)

2.4 CPE (carboxypeptidase E)

- Acidic, granule-matched (pH ≈ 5.5)
- **Zn^{2+} metalloenzyme** [UniProt P16870] — **do not add EDTA to a CPE reaction**
- Removes C-terminal K/R exposed by endoproteolysis

2.5 PAM (PHM → PAL)

- Requires **ascorbate + Cu^{2+} + O_2**; include catalase to suppress peroxide
- Bifunctional: PHM hydroxylates, PAL cleaves — two sequential domains [UniProt P19021]
- Inhibited by **EDTA, phenylglyoxal, diethyl pyrocarbonate** [UniProt P19021]
- Net mass change: **−58.00548 Da** (loss of Gly 57.02146 + 0.98402 amidation)

2.6 Furin — comparison arm

- pH optimum **6.0** [UniProt P09958], Ca^{2+}-dependent
- Tests whether cassette junctions are furin-accessible before the granule

CRITICAL — enzyme loading is not equal-mass

Relative specific activities on standard substrates [Yoon2022]:

enzyme	specific activity	substrate used
cathepsin L	**100%** (reference)	Z-Phe-Arg-AMC
cathepsin V	10%	Z-Phe-Arg-AMC
PC1/3	**1.5%**	pGlu-Arg-Thr-Lys-Arg-AMC
PC2	**0.07%**	pGlu-Arg-Thr-Lys-Arg-AMC

PC2 is roughly **1,400× less active than cathepsin L**. Loading equal enzyme mass across arms would under-sample the PC2 arm to the point of a false negative. **Normalise by activity units against the reference fluorogenic substrate, not by protein mass**, and report the units used.

2A. Materials — what to order

Enzymes, in the exact form the source used [Yoon2022]. Catalogue numbers are the ones cited in the paper; verify availability before ordering.

item	source in [Yoon2022]	cat. no.	notes
Human recombinant cathepsin L	R&D Systems	#952-CY-010	no preactivation needed
Human recombinant cathepsin V	R&D Systems	#1080-CY-010	**requires preactivation** (§2.3)
Human recombinant PC1/3	R&D Systems	#2810-SE-010	
Human recombinant PC2	R&D Systems	#6018-SE-010	must be 7B2-matured — confirm with vendor
Cathepsin H	R&D Systems	—	only for the coupled AMC assay (§2B)
CPE, PAM, furin	not used in [Yoon2022]	—	source separately; conditions in §2.4-2.6 are [UniProt]-derived

Fluorogenic substrates for activity normalisation:

substrate	supplier in [Yoon2022]	used for
Z-Phe-Arg-AMC	Bachem	cathepsin L and V titration
pERTKR-AMC (pGlu-Arg-Thr-Lys-Arg-AMC)	R&D Systems	PC1/3 and PC2 titration
Z-Tyr-Lys-Arg-AMC	GenScript	cross-enzyme specific-activity comparison
Z-Leu/Trp/Phe/Tyr/Val/Gly/Ala-Lys-Arg-AMC	GenScript	**P2 preference series — see §2B**

Reagents: octyl-β-glucopyranoside (Sigma-Aldrich), DTT (Promega), urea (Teknova), acetonitrile and TFA (Fisher Chemical), C18 SPE stage-tips (3M), BEH C18 packing (Waters), DMSO, BSA, MES, GEMSA (CPE inhibitor), ascorbate, $CuSO_4$, catalase.

Instrumentation used in the source: **Dionex UltiMate 3000 nano LC + Orbitrap Q-Exactive** (Thermo Fisher); 75 µm ID × 25 cm BEH C18 (1.7 µm) at 65 °C, 0.3 µL/min, 60 min 5-30% B then 15 min 30-85% B (B = 100% ACN in 0.1% TFA; A = 0.1% TFA in water). MS1 at 70,000 resolution over m/z 250-1500; MS2 at 17,500, HCD NCE 30 [verbatim Yoon2022]. Any comparable nano-LC-Orbitrap will do; these are the settings the reference conditions were established on.

2B. Activity normalisation — do this before the real experiment

Section 2's "CRITICAL" note says to normalise by activity, not mass. This is how.

1. **Titrate each enzyme against its fluorogenic substrate** — Z-Phe-Arg-AMC for the cathepsins, pERTKR-AMC for PC1/3 and PC2 — in that enzyme's own buffer (§2.1-2.3), 37 °C, monitoring AMC at **ex 360 / em 460 nm**.
2. Take the initial linear rate. Define **1 unit = 1 pmol AMC released per minute** at 37 °C under those conditions, and record it; the absolute definition matters less than using the same one for every arm.
3. **Load equal units, not equal ng**, into the PENK reactions. As a sanity anchor, the source's own MSP-MS loadings were cathepsin L 0.04, cathepsin V 0.16, PC1/3 3.6, and PC2 5.4 ng/µL — a ~135-fold mass spread that reflects exactly this normalisation.
4. **Optional but recommended — run the P2 series directly.** The central prediction of this study is a P2 effect, and it can be tested on pure chemistry before any PENK substrate is consumed: assay cathepsin L against Z-**F**-KR-AMC (the native P2) versus Z-**V**-KR-AMC (the engineered P2), with Z-**L**-KR-AMC and Z-**Y**-KR-AMC as preferred-set positives and Z-**G**-KR-AMC / Z-**A**-KR-AMC as poor-P2 negatives. In [Yoon2022] cathepsin L was most active on Z-L-KR-AMC, Z-F-KR-AMC, and Z-Y-KR-AMC. The Phe-vs-Val rate ratio measured here is the cleanest possible statement of the predicted defect, uncontaminated by folding or accessibility. Note the cathepsins need a **coupled assay with cathepsin H** to liberate AMC from Z-Xaa-Lys-Arg-AMC (add cathepsin H at 0.5 ng/µL in 50 mM MES to bring pH to 6.5, then 30 min at 37 °C); PC1/3 and PC2 give the same result with or without it [Yoon2022]. Fluorogenic assay buffer for the cathepsins: 5 mM citrate-phosphate pH 5.5, 4 mM DTT, 1 mM EDTA, 50 mM NaCl, 1% DMSO, 60 µM substrate, plus 0.001% BSA for cathepsin L; for PC1/3 and PC2: 5 mM Na-acetate pH 5.5, 5 mM $CaCl_2$, 50 mM NaCl [verbatim Yoon2022].

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3. Buffer incompatibilities — read before combining enzymes

These cannot all go in one tube:

1. **Cathepsin L buffer contains 1 mM EDTA. CPE and PAM are metalloenzymes.** EDTA will inhibit both. Cathepsin L and CPE/PAM **must be staged sequentially with a buffer exchange**, never co-incubated.
2. **DTT (5 mM, cathepsin L) will reduce disulfides.** Native PENK has cysteines in the N-terminal domain; in Stage 2 this will unfold the proprotein and destroy the conformational context the stage exists to test. Run the Stage-2 cathepsin arm as a deliberate denatured-accessibility comparison, not as the physiological arm.
3. **PC buffers use Na-acetate; cathepsin uses citrate phosphate.** Citrate chelates Ca^{2+}. Do not substitute one for the other.

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4. Order of addition

Stage 1 — junction peptides (establish conditions)

1. Reconstitute each peptide pool to 0.5 µM per peptide final (matching the library concentration in [Yoon2022]).
2. Pre-warm substrate + buffer to 37 °C.
3. Remove the **t = 0 aliquot before adding enzyme.**
4. Add enzyme; start timer.
5. Sample at **0, 15, 30, 60, 120 min** (the published assay reads 30 and 60 min).
6. **Quench: transfer a 10 µL aliquot into 60 µL of 8 M urea, vortex, store at −70 °C** [verbatim Yoon2022]. This gives **6.86 M urea final** in 70 µL.

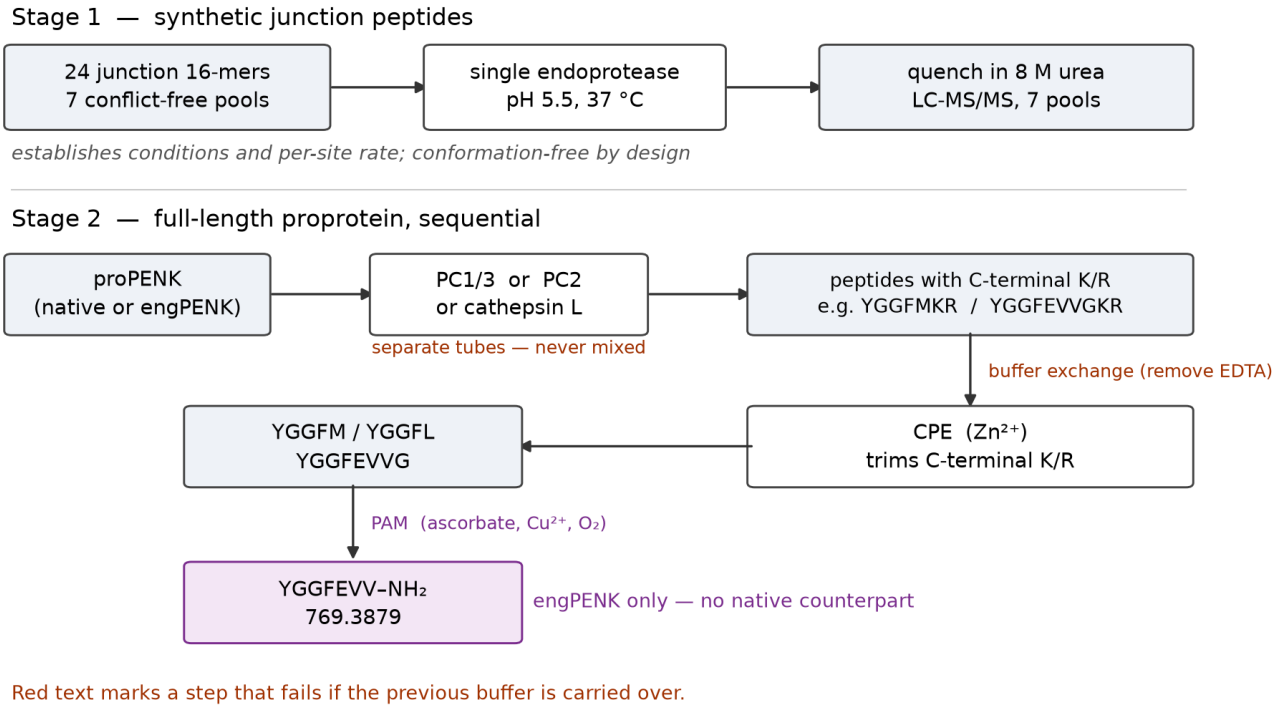
7. Acidify with 40 µL 1% TFA; desalt on C18 SPE stage-tips, eluting sequentially with 50% then 80% ACN in 0.1% TFA; dry in a SpeedVac; store at −70 °C [verbatim Yoon2022].
8. Resuspend in 40 µL 0.1% TFA for LC-MS/MS.

Stage 2 — full-length proprotein (confirm in context)

1. Endoprotease first (PC1/3 **or** PC2 **or** cathepsin L — separate tubes, never mixed).
2. Quench / buffer exchange.
3. **Then** CPE (removes exposed C-terminal K/R).
4. Buffer exchange (remove any EDTA).
5. **Then** PAM (+ascorbate, Cu^{2+}, catalase) — engPENK arm especially.

Order matters: CPE cannot act until endoproteolysis exposes a C-terminal basic residue, and PAM cannot act until CPE has exposed the Gly.

Figure 3 — Staged cascade. Order of addition for both stages. Red annotations mark steps that fail if the preceding buffer is carried over.



5. Controls

control	purpose	expected result
t = 0, no enzyme	baseline	intact substrate only
Heat/urea-denatured enzyme	rules out non-enzymatic hydrolysis	no products [Yoon2022 used urea-denatured enzyme as the "0" condition]
Native arm	assay validation	Met-enkephalin (574.2330) must appear
STD-CPE (`YGGFMKR`)	CPE activity, independent of endoprotease	trims to `YGGFM`
GEMSA	CPE-specific inhibitor	blocks the trim step only
EDTA add-back	confirms metal dependence	abolishes CPE and PAM, not cathepsin L
No-ascorbate PAM	confirms PAM cofactor dependence	Gly-extended peptide persists
Furin arm	pre-granule accessibility	reveals whether junctions are cut early

6. Peptide pooling — why 7 pools and not 1

All seven engPENK cassettes release the **identical** C-terminal product `YGGFEVVG` (827.3934). Pooling peptides whose products are isobaric makes the MS assignment ambiguous. A mass-conflict graph was built over every species each peptide can generate (intact, both products, CPE-trimmed N-product, and +1/+2 Met oxidation) at **20 ppm**, then coloured into the minimum number of conflict-free pools.

- **7 pools, 3-4 peptides each, zero intra-pool conflicts.**
- Splitting simply by construct **fails** — 31 conflicting pairs in the engPENK arm alone.
- **Pool 6 caveat:** N099 (1956.0272) and N106 (1956.9757) are baseline-resolved at MS1, but N099's M+1 isotope peak lies 28 ppm from N106's monoisotopic peak and the two have identical GRAVY (−0.850), so they may co-elute. **Require isotope-envelope deconvolution and confirm both by MS2.**

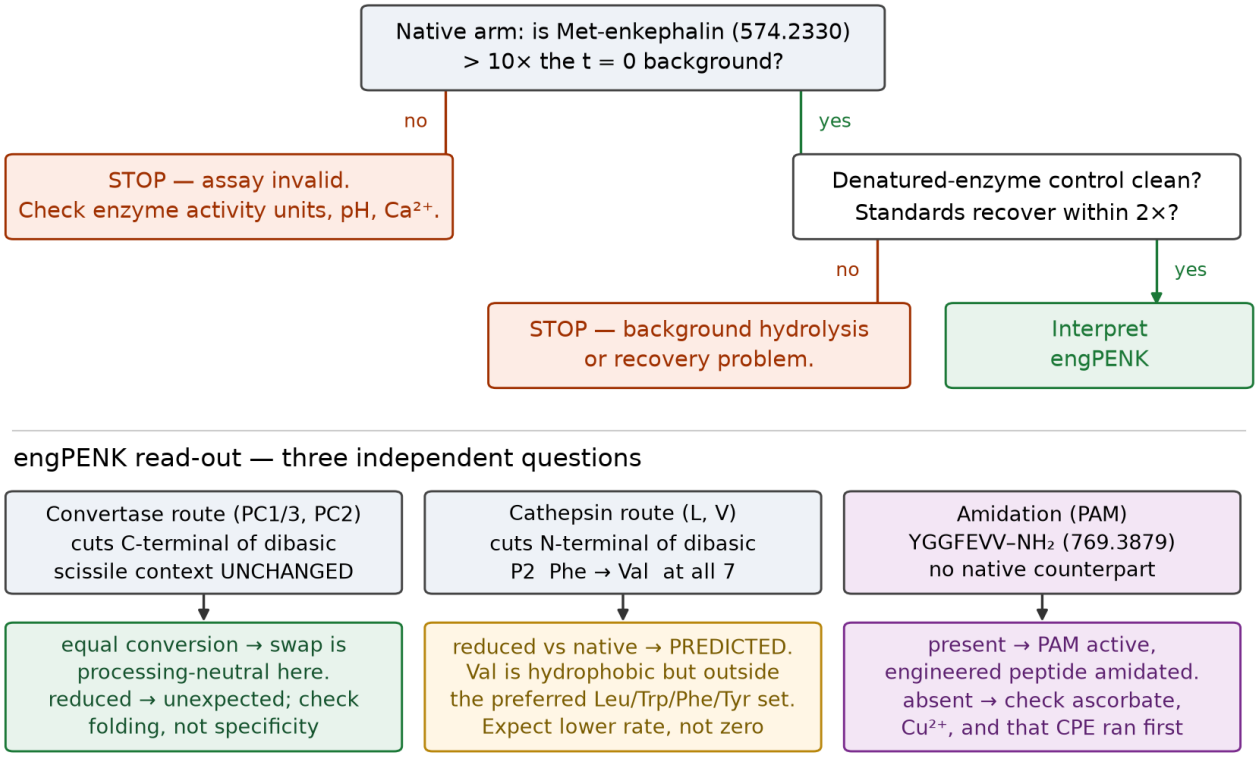
7. Readout and acceptance criteria

Accept the run if: - native arm produces Met-enkephalin (574.2330) above 10× the t=0 background, and - denatured-enzyme control produces no products, and - the internal standards recover within 2-fold across pools.

Then, and only then, interpret engPENK.

Primary comparison: fractional conversion at each paired junction, native vs engPENK, at matched enzyme activity units and matched time.

Figure 4 — Decision tree. Gate the run on the native arm first; only then interpret engPENK against the three independent questions.



Every comparison is made at matched enzyme activity units and matched time, native vs engPENK.

8. Known limitations

1. **Shennan1995 pH/Ca²⁺ values describe zymogen maturation, not mature-enzyme turnover.** pro-PC2 matures optimally at pH 5.5-6.0 with Ca^{2+} $K_{0.5}$ = 2-4 mM; pro-PC3 matures at pH 7.0-8.0 without added Ca^{2+}. These describe where in the secretory pathway each zymogen activates — they are **not** turnover optima for the mature enzyme and must not be used as such. The assay pH above comes from [Yoon2022], which used mature enzyme.
2. **A cell-free reconstitution cannot reproduce granule crowding, the true lumenal pH gradient, or the local enzyme concentrations of a secretory granule.** Rate ratios between enzymes will differ from the cell.
3. **Stage 1 removes conformational context by design.** A junction that is cleaved as a 16-mer may be buried in the folded proprotein. This is precisely why Stage 2 exists; do not report Stage 1 rates as proprotein rates.
4. **Dupuy2003 is cited from its abstract only** (paywalled). The PC1-vs-PC2 site assignments it provides should be confirmed against the full text before publication.

9. Methods (for the manuscript)

Sequence analysis. Human proenkephalin (UniProt P01210, 267 aa) and the engineered construct engPENK (288 aa) were analysed in Biopython. Signal-peptide boundaries were taken from the UniProt P01210 SIGNAL annotation (residues 1-24). Candidate endoproteolytic sites were enumerated as all dibasic (KR, KK, RR, RK) and licensed monobasic (R/K with a basic residue at P4 or P6) positions and assigned to three tiers: T1 canonical dibasic (KR/RR), T2 slow dibasic (KK), T3 licensed monobasic. Scissile-bond contexts were computed separately for the two mechanistic routes, since they cut on opposite sides of the dibasic [Yoon2022].

Mass calculation. Monoisotopic and average masses were computed with Pyteomics (`pyteomics.mass`). C-terminal amidation was modelled as -0.98402 Da; the PAM reaction as a net -58.00548 Da from the Gly-extended precursor (loss of the Gly residue, 57.02146 Da, plus the amidation offset). Methionine oxidation was modelled as +15.99491 Da per oxidation, enumerated at one and two oxidations. The mass engine was validated against ten independently stored reference species and agreed to better than 0.05 ppm.

Peptide pooling. Junction peptides were pooled for multiplexed LC-MS/MS by graph colouring. An undirected conflict graph was constructed over the 24 peptides in which an edge joins any pair that could generate an indistinguishable species; the species set per peptide comprised the intact peptide, both endoproteolytic products, the carboxypeptidase-trimmed N-product, and the +1/+2 methionine-oxidation variants of each. Two species conflict if they are sequence-identical or if their neutral monoisotopic masses fall within 20 ppm. The graph (50 edges) was coloured by randomised-restart greedy colouring under a pool-count cap, minimising the maximum pool size; the accepted solution uses 7 pools of 3-4 peptides with zero intra-pool conflicts.

Reference retrieval. Primary literature was retrieved through NCBI E-utilities and the PubMed Central open-access corpus. Enzyme cofactor requirements, pH optima, and activity-regulation annotations were taken from UniProt (release retrieved during this analysis). Buffer compositions reproduced verbatim from a primary source are marked as such in the protocol; values from database annotation are marked separately.

Software. Python 3, Biopython, Pyteomics, pandas, NumPy, Matplotlib.

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10. References

1. **Yoon MC, Ames J, Mosier C, Jiang Z, Podvin S, O'Donoghue AJ, Hook V** (2022). Distinct dibasic cleavage specificities of neuropeptide-producing cathepsin L and cathepsin V cysteine proteases compared to PC1/3 and PC2 serine proteases. ACS Chem Neurosci. doi:10.1021/acschemneuro.1c00653 · PMID 34986304 · PMC9070308 · **full text** Source of: PC1/3, PC2 and cathepsin L/V assay buffers (verbatim); the Leu/Trp/Phe/Tyr P2 preference of cathepsin L; relative specific activities; the finding that cathepsins cut N-terminal to the dibasic while PC1/3 and PC2 cut C-terminal.

2. **Dupuy A, Lindberg I, Zhou Y, Akil H, Lazure C, Chretien M, Seidah NG, Day R** (2003). Cleavage of recombinant proenkephalin and blockade mutants by prohormone convertases 1 and 2: an in vitro specificity study. J Neurochem 87:868-878. doi:10.1046/j.1471-4159.2003.02043.x · PMID 12716426 · **abstract only (paywalled)** Source of: PC1-vs-PC2 site assignment on proenkephalin itself; PC2 uniquely cleaves the RR-V and KK-M sites; PC1 excludes large aliphatic residues at P1'; processing order is non-obligatory. Confirm against full text before publication.

3. **Shennan KIJ, Taylor NA, Jermany JL, Matthews G, Docherty K** (1995). Differences in pH optima and calcium requirements for maturation of the prohormone convertases PC2 and PC3. J Biol Chem 270:1402-1407. doi:10.1074/jbc.270.3.1402 · PMID 7836406 · **abstract only (paywalled)** Source of: pro-PC2 maturation optimum pH 5.5-6.0, Ca2+ K0.5 = 2-4 mM; pro-PC3 maturation optimum pH 7.0-8.0, Ca2+-independent. **These are zymogen-maturation parameters, not mature-enzyme turnover optima — see Limitations 8.1.**

4. **Blanco EH, Peinado JR, Martin MG, Lindberg I** (2014). Biochemical and cell biological properties of the human prohormone convertase 1/3 Ser357Gly mutation: a PC1/3 hypermorph. Endocrinology 155:3434-3447. doi:10.1210/en.2013-2151 · PMID 24932808 · PMC4138575 Source of: wild-type human PC1/3 is Ca2+-dependent with an acidic pH optimum; the S357G hypermorph shifts toward neutral pH and reduced Ca2+ dependence. Relevant only if a variant enzyme is sourced.

5. **UniProt Consortium** (2026). UniProt: the universal protein knowledgebase. Nucleic Acids Res (database issue). doi:10.1093/nar/gkae1010 Entries used: P01210 (PENK), P29120 (PCSK1/

PC1/3), P16519 (PCSK2/PC2), P05408 (SCG5/7B2), P16870 (CPE), P19021 (PAM), P09958 (FURIN), P07711 (CTSL), P08473 (MME/neprilysin), P12821 (ACE), P15144 (ANPEP). Source of: cofactor requirements, annotated pH optima, activity-regulation notes (PAM inhibited by EDTA / phenylglyoxal / diethyl pyrocarbonate; PC2's obligate 7B2 chaperone), and the P01210 signal-peptide boundary.